

Constrained-Path Auxiliary-Field Quantum Monte Carlo

Theory, Algorithm, and the `pyqmc` Implementation

CPQMC validation platform — `codex-CPQMC/pyqmc`

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Abstract

This note derives the constrained-path auxiliary-field quantum Monte Carlo (CP-AFQMC / CPMC) algorithm of Zhang and Krakauer as implemented in `pyqmc/cpqmc.py`, the Python port of the validation platform. We give the ground-state projection, the Trotter–Suzuki split, the discrete Hubbard–Stratonovich transformation actually used in the code, importance sampling, the constrained-path approximation, the mixed and back-propagated estimators, and the observables (energy, equal-time correlations, singlet pairing, magnetization). We close with the adaptive self-consistent trial wavefunction (Phase 6), an algorithm flowchart, and a line-by-line map from the equations to the code.

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1 Model

The platform targets density–density lattice fermion Hamiltonians,

$$\hat{H} = \hat{K} + \hat{V}, \quad \hat{K} = \sum_{ij,\sigma} K_{ij} \hat{c}_{i\sigma}^\dagger \hat{c}_{j\sigma}, \quad \hat{V} = \sum_{(as_a, bs_b)} V_{ab} \hat{n}_{as_a} \hat{n}_{bs_b}, \quad (1)$$

with $\hat{n}_{a\sigma} = \hat{c}_{a\sigma}^\dagger \hat{c}_{a\sigma}$. The single-band Hubbard model has $K_{ij} = -t$ on nearest neighbours and a single on-site term $V = U$, $\hat{V} = U \sum_i \hat{n}_{i\uparrow} \hat{n}_{i\downarrow}$. The two-orbital altermagnet (the PRL LE20050 model) uses the anisotropic two-orbital hopping K built in `altermagnet_ed.build_hopping`, plus on-site u_{xx} , inter-orbital u_{xy} , and neighbour v (intra-orbital $+v$, inter-orbital $-v$). All three reduce to the general density–density form of Eq. (1), which is why a single propagation/estimator engine covers every channel.

Listing 1: Interaction stored as a generic density–density term list (`LatticeModel.build_terms`).

```
# on-site Hubbard U on every (orbital-)site: n_{i,up} n_{i,dn}
self.terms.append((i, 0, i, 1, e1, e2, cf)) # (a, s_a, b, s_b, HS factors)
self.pot_terms.append((i, 0, i, 1, U)) # (a, s_a, b, s_b, V)
```

2 Ground-state projection

CPMC obtains the ground state by imaginary-time projection from a trial state $|\psi_T\rangle$ non-orthogonal to it,

$$|\psi_0\rangle \propto \lim_{\beta \rightarrow \infty} e^{-\beta \hat{H}} |\psi_T\rangle = \lim_{n \rightarrow \infty} \left(e^{-\Delta\tau \hat{H}} \right)^n |\psi_T\rangle, \quad \beta = n \Delta\tau. \quad (2)$$

The trial is taken as the free-electron ground state of $\hat{K}$ (the lowest $N_\uparrow / N_\downarrow$ eigenvectors), a single Slater determinant per spin.

Listing 2: Trial = free-electron ground state (`CPMC.__init__`).

```
w, vecs = np.linalg.eigh(self.K) # single-particle eigenstates
self.trial = TrialWF(vecs[:, :nup].copy(), vecs[:, :ndn].copy())
```

3 Trotter–Suzuki decomposition

For small $\Delta\tau$ the symmetric (second-order) split separates kinetic and interaction parts,

$$e^{-\Delta\tau \hat{H}} = e^{-\frac{\Delta\tau}{2} \hat{K}} e^{-\Delta\tau \hat{V}} e^{-\frac{\Delta\tau}{2} \hat{K}} + \mathcal{O}(\Delta\tau^3). \quad (3)$$

The one-body factor $B_K \equiv e^{-\frac{\Delta\tau}{2} \hat{K}}$ acts on a Slater determinant Φ (an $N_s \times N_\sigma$ matrix of orbitals) by left multiplication, $\Phi \rightarrow e^{-\frac{\Delta\tau}{2} K} \Phi$, preserving the determinant form (Thouless’ theorem).

Listing 3: Half-step kinetic propagator (`LatticeModel`).

```
self.expK = vecs @ np.diag(np.exp(-0.5 * dt * w)) @ vecs.T # e^{-dt K/2}
```

4 Discrete Hubbard–Stratonovich transformation

The interaction $e^{-\Delta\tau V \hat{n}_a \hat{n}_b}$ is quartic in the fermion operators; it is decoupled into a sum over an Ising-like auxiliary field $s = \pm 1$ so that each term becomes a one-body operator. For each density–density pair the code uses the discrete Hirsch decoupling. Writing $\cosh \gamma = e^{\frac{\Delta\tau}{2}|V|}$:

- **Repulsive** ($V > 0$): the *spin* channel,

$$e^{-\Delta\tau V \hat{n}_a \hat{n}_b} = e^{-\frac{\Delta\tau V}{2}(\hat{n}_a + \hat{n}_b)} \frac{1}{2} \sum_{s=\pm 1} e^{\gamma s(\hat{n}_a - \hat{n}_b)}, \quad (4)$$

implemented as operator-1 on (a, s_a) with factor $e^{\gamma s - \Delta\tau V/2}$ and operator-2 on (b, s_b) with factor $e^{-\gamma s - \Delta\tau V/2}$.

- **Attractive** ($V < 0$): the *charge* channel; both operators take $e^{\gamma s - \Delta\tau V/2}$ and the field carries a weight $e^{-\gamma s + \Delta\tau V/2}$.

For the on-site Hubbard term $a = b = i$, $s_a = \uparrow$, $s_b = \downarrow$, Eq. (4) is the textbook Hirsch transform with $\cosh \gamma = e^{\Delta\tau U/2}$.

Listing 4: Discrete HS factors (`LatticeModel.hs`).

```
a = np.arccosh(np.exp(0.5 * dt * abs(V))) # gamma, cosh(g)=e^{dt|V|/2}
if V > 0: # repulsive: spin channel
    e1 = np.array([np.exp(a - 0.5*dt*V), np.exp(-a - 0.5*dt*V)])
    e2 = np.array([np.exp(-a - 0.5*dt*V), np.exp(a - 0.5*dt*V)])
    cf = np.array([1.0, 1.0])
else: # attractive: charge channel
    e1 = np.array([np.exp(a - 0.5*dt*V), np.exp(-a - 0.5*dt*V)]); e2 = e1.copy()
    cf = np.array([np.exp(-a + 0.5*dt*V), np.exp(a + 0.5*dt*V)])
```

After the transformation a full time step is a *sum over auxiliary-field configurations* of one-body propagators:

$$e^{-\Delta\tau \hat{H}} \Phi = \sum_{\{s\}} P(\{s\}) B_K B_V(\{s\}) B_K \Phi, \quad (5)$$

each $B_V(\{s\})$ being diagonal scalings of the determinant rows. Because a one-body operator maps a Slater determinant to a Slater determinant, the projection (2) becomes an open-ended random walk in the manifold of determinants.

5 Random walk, importance sampling, constrained path

A population of *walkers* $\{|\phi_i\rangle, w_i\}$ represents the projected state stochastically, $|\psi\rangle \approx \sum_i w_i |\phi_i\rangle / \langle \psi_T | \phi_i \rangle$. Naive sampling of $\{s\}$ has a large variance; the code uses *importance sampling*, choosing each field with probability proportional to the overlap ratio with the trial,

$$p_s \propto c_s \max\left(\frac{\langle \psi_T | \phi^{(s)} \rangle}{\langle \psi_T | \phi \rangle}, 0\right), \quad (6)$$

where $\phi^{(s)}$ is the determinant after applying the field- s one-body factor and c_s is the HS weight (`cf`). The walker weight is multiplied by the normalization $w_i \rightarrow w_i \cdot \frac{1}{2} \sum_s c_s R_s$.

Constrained-path approximation. The $\max(\cdot, 0)$ in Eq. (6) is the constrained path: walkers whose overlap with the trial would turn non-positive are removed. This eliminates the fermion sign problem (the walk stays in the half-space $\langle \psi_T | \phi \rangle > 0$) at the cost of a bias that vanishes as $|\psi_T\rangle \rightarrow \psi_0$.

Listing 5: One importance-sampled, constrained step (`Propagator.step`).

```
for (a, sa, b, sb, e1, e2, cf) in m.terms:
    ratios = np.empty(2)
    for k in (0, 1): # HS field s = +1 (k=0), -1 (k=1)
        R = self._term_ratio(pu, pd, ov_up, ov_dn, a, sa, b, sb, e1[k], e2[k])
        ratios[k] = cf[k] * max(R, 0.0) # constrained path: clip R <= 0
    tot = ratios.sum()
    if tot <= 0:
        walkers.w[i] = 0.0; killed = True; break
    k = self.rng.choice(2, p=ratios / tot) # importance sampling
    # ... apply chosen one-body factor to rows a, b ...
    walkers.w[i] *= 0.5 * tot
```

Stabilization and population control. Repeated B_K multiplication makes the orbital columns numerically collinear; the code periodically re-orthonormalizes each determinant by a QR factorization (the determinant is invariant up to the discarded R). A comb-resampling population-control step keeps the walker count fixed while pruning small weights and branching large ones.

6 Estimators

6.1 Mixed estimator

For an operator $\hat{O}$ the mixed estimator is

$$\langle \hat{O} \rangle_{\text{mix}} = \frac{\sum_i w_i \frac{\langle \psi_T | \hat{O} | \phi_i \rangle}{\langle \psi_T | \phi_i \rangle}}{\sum_i w_i}. \quad (7)$$

Matrix elements between Slater determinants reduce, by the generalized Wick theorem, to the single-particle Green's function

$$G_{ij}^\sigma = \frac{\langle \psi_T | \hat{c}_{i\sigma}^\dagger \hat{c}_{j\sigma} | \phi \rangle}{\langle \psi_T | \phi \rangle} = \left[\Phi (\Psi_T^\dagger \Phi)^{-1} \Psi_T^\dagger \right]_{ji}. \quad (8)$$

The energy is exact for $\hat{H}$ (zero-variance principle) at $|\psi_T\rangle = \psi_0$; operators not commuting with $\hat{H}$ carry an $O(\text{systematic})$ bias.

Listing 6: Green's function and mixed energy (`Estimators.energy`).

```
def _green(left, phi):
    A = left.T @ phi
    return phi @ np.linalg.solve(A, left.T) # G_ij = <c^+_j c_i>
...
ek = np.sum(m.K * Gu.T) + np.sum(m.K * Gd.T) # kinetic
ev += V * (Gs[a,a]*Gs[b,b] - Gs[a,b]*Gs[b,a]) # same-spin Wick (exchange)
ev += V * G[sa][a,a]*G[sb][b,b] # opposite-spin (no exchange)
```

6.2 Back-propagation

The mixed bias for $\hat{O} \neq f(\hat{H})$ is removed by back-propagation: the bra $|\psi_T\rangle$ is itself projected for m steps using the *recorded* auxiliary fields, $\langle\psi_T| \rightarrow \langle\psi_T| \prod B_l^\dagger$, giving a pure (left/right both projected) estimator

$$\langle\hat{O}\rangle_{\text{bp}} = \frac{\sum_i w_i \langle\psi_T| \left(\prod_l B_l \right)^\dagger \hat{O} |\phi_i\rangle}{\sum_i w_i \langle\psi_T| \left(\prod_l B_l \right)^\dagger |\phi_i\rangle}. \quad (9)$$

The constrained-path bias from the trial *node* remains; back-propagation removes only the mixed-estimator part.

Listing 7: Back-propagated bra (`Estimators.bp_bra`).

```
Lu = tr.up.copy(); Ld = tr.dn.copy()
for s in range(bp - 1, -1, -1): # replay recorded fields, reversed
    Lu = m.expK @ Lu; Ld = m.expK @ Ld
    for (a, sa, b, sb, e1, e2, cf), k in zip(m.terms, ch):
        # apply the recorded field-k factors to the bra
    Lu = m.expK @ Lu; Ld = m.expK @ Ld
```

6.3 Two-body observables

Equal-time correlations follow from G by Wick's theorem (spin-separated determinants, so no cross-spin contraction):

$$\langle\hat{n}_i\hat{n}_j\rangle = (n_i^\uparrow + n_i^\downarrow)(n_j^\uparrow + n_j^\downarrow) + \sum_\sigma (\delta_{ij}n_i^\sigma - G_{ij}^\sigma G_{ji}^\sigma), \quad (10)$$

$$\langle S_i^z S_j^z \rangle = \frac{1}{4} \left[(n_i^\uparrow - n_i^\downarrow)(n_j^\uparrow - n_j^\downarrow) + \sum_\sigma (\delta_{ij}n_i^\sigma - G_{ij}^\sigma G_{ji}^\sigma) \right]. \quad (11)$$

The singlet pairing structure factor for symmetry α with bond form factor f_α and pair operator $\hat{\Delta}_\alpha(m) = \sum_\delta f_\alpha(\delta) \hat{c}_{m\uparrow} \hat{c}_{m+\delta\downarrow}$ is

$$S_\alpha = \sum_{m,n} \langle \hat{\Delta}_\alpha^\dagger(m) \hat{\Delta}_\alpha(n) \rangle = \sum_{m,n} G_{mn}^\dagger (B_\alpha G^\dagger B_\alpha^\top)_{mn}, \quad (B_\alpha)_{m,m+\delta} = f_\alpha(\delta), \quad (12)$$

exact at $U = 0$ and used for the s - and $d_{x^2-y^2}$ -wave channels (`Estimators.pairmag_block`).

6.4 Dynamic (imaginary-time-displaced) correlations

Unequal-time correlations $C_O(\tau) = \langle\psi_0| \hat{O} e^{-\tau(\hat{H}-E_0)} \hat{O} |\psi_0\rangle$ and the static susceptibility $\chi_O = \int_0^\infty C_O(\tau) d\tau$ follow from the imaginary-time-displaced single-particle Green's functions. With the one-body slice propagator $B_{(\ell)} = B_\ell \cdots B_1$ (built from the recorded auxiliary fields), and $g_{ij}^\sigma = \langle \hat{c}_{i\sigma}^\dagger \hat{c}_{j\sigma} \rangle$ the back-propagated equal-time function, the particle and hole displaced functions and the displaced density are

$$P^\sigma(\tau) = B_{(\ell)}(I - g^{\sigma\top}), \quad H^\sigma(\tau) = B_{(\ell)}^{-\top} g^\sigma, \quad g^\sigma(\tau) = B_{(\ell)}^{-\top} g^\sigma B_{(\ell)}^\top, \quad (13)$$

using $\hat{c}_i(\tau) = \sum_j [B_{(\ell)}]_{ij} \hat{c}_j$ and $\hat{c}_i^\dagger(\tau) = \sum_k [B_{(\ell)}^{-\top}]_{ik} \hat{c}_k^\dagger$. The same-spin density-density correlation is then $\langle \hat{n}_{i\sigma}(\tau) \hat{n}_{j\sigma}(0) \rangle_c = H_{ij}^\sigma(\tau) P_{ij}^\sigma(\tau)$, from which the staggered spin correlation $C(\tau) = \sum_{ij} (-1)^{r_i+r_j} \langle S_i^z(\tau) S_j^z(0) \rangle$ is assembled. (The transposes in Eq. (13) are invisible at $U = 0$, where B and g are symmetric, but essential once interacting — the code is validated against the ED Lehmann representation $C(\tau) = \sum_n |\langle n | \hat{O} | \psi_0 \rangle|^2 e^{-(E_n - E_0)\tau}$.)

Listing 8: Time-displaced GFs (`Estimators.chi_block`).

```
nu = np.diag(Bui.T @ gu @ Bu.T) # density n^up_i(tau)
Hu = Bui.T @ gu; Pu = Bu @ ImguT # hole / particle displaced GF
conn = 0.25*(phase @ (Hu*Pu) @ phase + phase @ (Hd*Pd) @ phase) # same-spin
```

7 Adaptive self-consistent trial (Phase 6)

Because the constrained-path bias is controlled entirely by the quality of $|\psi_T\rangle$, the code can replace the fixed free-electron trial by a self-consistent one. Each generation it forms the weighted ensemble one-body density matrix, symmetrizes it, damps it toward the current trial projector, and takes the highest-occupation natural orbitals as the new trial:

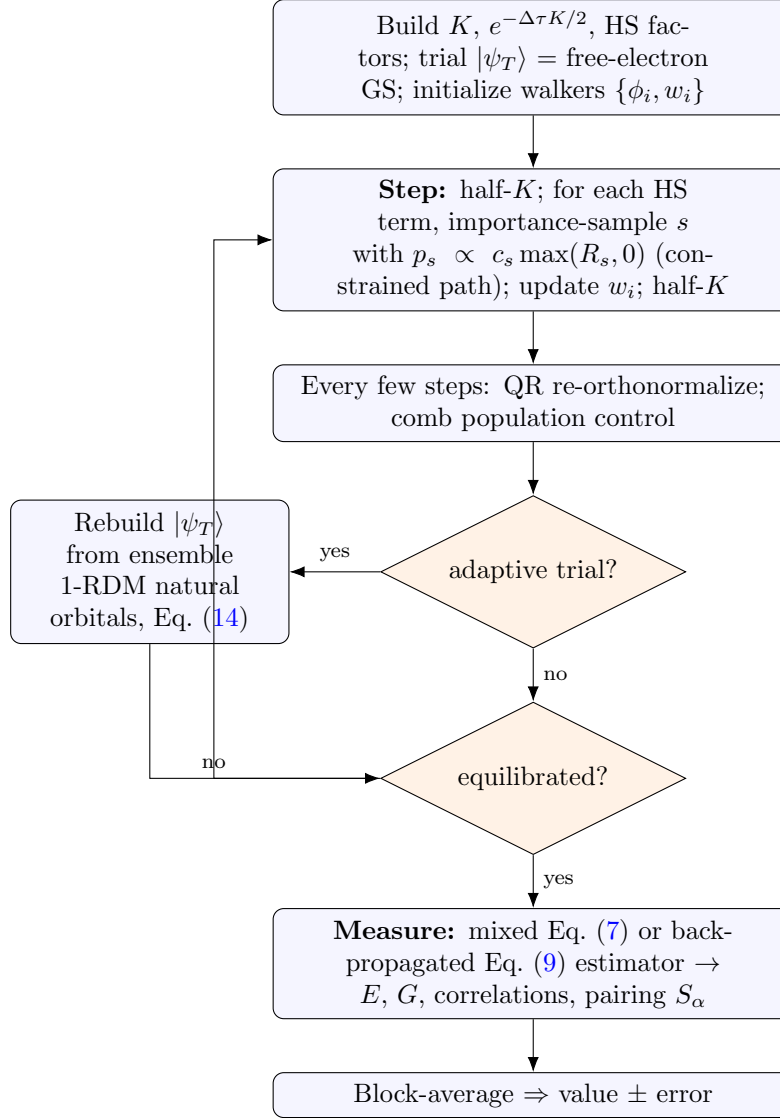
$$\rho^\sigma = \frac{\sum_i w_i G_i^\sigma}{\sum_i w_i}, \quad \tilde{\rho}^\sigma = \mu \Psi_T^\sigma \Psi_T^{\sigma\top} + (1 - \mu) \frac{1}{2}(\rho^\sigma + \rho^{\sigma\top}), \quad \Psi_T^\sigma \leftarrow \text{top-}N_\sigma \text{ eigvecs}(\tilde{\rho}^\sigma). \quad (14)$$

This moves the constraint boundary toward the true ground state. It is *non-variational* (the trial is no longer a fixed external constraint), so its benefit is assessed empirically against ED; **fixed** remains the validated default.

Listing 9: Natural-orbital self-consistent update (`TrialWF.update`).

```
ru += w * Gu; rd += w * Gd; W += w # ensemble 1-RDM
ru = 0.5*(ru+ru.T); rd = 0.5*(rd+rd.T) # symmetrize (mixed RDM not Herm.)
ru = self.mix*Pu + (1-self.mix)*ru # damp toward current trial
_, Vu = np.linalg.eigh(ru); self.up = Vu[:, -self.nup:].copy() # natural orbitals
```

8 Algorithm



9 Theory $\leftrightarrow$ code map

Concept	Equation	Code (pyqmc/cpqmc.py)
Model / interaction list	(1)	LatticeModel._build_terms
Half-step kinetic B_K	(3)	LatticeModel.expK
Discrete HS factors	(4)	LatticeModel.hs
Importance + constraint	(6)	Propagator.step
Field recording (for BP)	(9)	Propagator.step_record
QR stabilize / pop control	—	WalkerEnsemble.reorthogonalize/pop_control
Green’s function	(8)	_green
Mixed energy	(7)	Estimators.energy
Back-propagated bra	(9)	Estimators.bp_bra, energy_bp
Pairing / magnetization	(12)	Estimators.pairmag_block
Dynamic susceptibility	(13)	Estimators.chi_block, run_bp_chi
Adaptive trial	(14)	TrialWF.update

Validation status. Energy agrees with ED on all three interaction channels (u_{xx}, u_{xy}, v); equal-time correlations and the pairing/spin estimators are exact at $U = 0$ and carry the expected constrained-path bias when interacting; the adaptive trial reduces that bias toward ED. See docs/VALIDATION.md for the quantitative log.